Subramanian Veerappan

Process Design Engineer | Copenhagen, Denmark
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Profile

Process engineer with a BTech in Mechanical Engineering and an MSc in Energy Engineering (Thermal and Process Engineering) from Aalborg University. I spent more than a year on two pyrolysis plants in Denmark, one in the FEED phase and one in operation, working with process models, P&IDs, equipment datasheets and plant data in Python. Outside work I have been building a portfolio of equipment sizing studies and a sample P&ID. I want to grow into energy and heat systems such as district heating and geothermal, and I learn new tools and standards quickly.

Core Skills

- **Process design and documentation:** heat and mass balances, BFD and PFD development, P&ID review, FEED deliverables, datasheets and specifications, stream tables; AutoCAD Plant 3D, Visio, SharePoint and ERP
- **Process safety:** PSV revalidation support, nitrogen blanketing calculations; knowledge of HAZOP, FMEA, FTA, What If and Checklist methods
- **Simulation and data analysis:** DWSIM, Aspen HYSYS, Aspen Flare Analyzer, FluidFlow; Python (SCADA time series, automated reporting), MATLAB, Excel
- **Equipment sizing:** pump, control valve, PSV and flare (API 520/521/527), hydraulics and pressure drop; heat exchanger (TEMA) and compressor sizing from course work (compressor in progress)

Experience

Process Design Engineer *Aug 2024 – Sep 2025*
MAKEEN Energy EnviroTech, Fredericia and Randers, Denmark (internship, then contract)
Fredericia, greenfield pyrolysis plant (FEED phase)

- Built the steady state DWSIM process model and stream tables used in design reviews
- Developed the BFD into a PFD and reviewed early P&IDs, coordinating stakeholder comments
- Prepared equipment datasheets and specifications for the FEED deliverables
- Kept project documents under version control in SharePoint and ERP

Randers, operational pyrolysis plant

- Supported PSV revalidation and nitrogen blanketing calculations
- Analysed three months of SCADA data, millions of rows, in Python to find the causes of downtime
- Built a reusable Python reporting workflow from it and presented the findings, with a suggestion for when the reactor could be stopped
- Went on site visits and reviewed field documentation; worked with a senior engineer to document carbon black buildup in vent and reactor outlet pipes for root cause analysis, using reports and videos
- Acted as the link between the Danish process team and MAKEEN's drawing team in India

Selected Engineering Projects

Independent Process Engineering Portfolio *Oct 2025 – Present*
Self initiated work, separate from MAKEEN: pump, control valve, PSV and flare sizing and a sample P&ID, shown in my online portfolio.

MSc Projects, Aalborg University

- Built two models in MATLAB for a high temperature PEM fuel cell in micro CHP, with a

sensitivity analysis *Jul 2023 – Jan 2024*

- Wrote the MATLAB code for a three stage adiabatic methanol reactor with intercooling: balances, kinetics and pressure drop, for a 1000 kg/day target *Feb 2022 – Jun 2022*

Education

MSc Energy Engineering (Thermal and Process Engineering) *Sep 2021 – Jan 2024*
Aalborg University, Denmark

BTech Mechanical Engineering (Energy specialization) *Sep 2016 – Jun 2020*
Shiv Nadar University, India

Volunteer Experience

Camp Committee Coordinator, Dansk ICYE *Mar 2025 – Present*

- Coordinate planning and execution of national youth camps with a diverse international committee
- Represented Dansk ICYE at EU level exchange camps to share practices across organisations

Additional Information

Languages: English (Fluent), Tamil (Native), Hindi, Telugu, Malayalam (Working Proficiency), Danish (2 of 6 modules completed)

Online certifications: PSV and flare sizing, control valve sizing, pump sizing, heat exchanger design, compressor sizing (in progress), AutoCAD P&ID drafting, Lean Six Sigma White and Yellow Belt

Availability: Immediate | Open to relocation and periodic travel

References: Available on request

Further technical details, engineering studies and selected project work are available in the online portfolio: subbu1310.github.io